Test Case Document					
Group Number - 29					MIhunan V. 214134H
Test Identifier	**Test Summery**	**Test Description**	**Precondition**	**Test Steps**	**Expected Result**
UI TEST CASES – USER ROLE					
TC_UI_USER_001	User sees Plant List	Verify user view of plants	User logged in, plants exist	1. Navigate to /plants	Plant table visible No edit/delete buttons
TC_UI_USER_002	User does not see "Add Plant" button	Verify no creation access	User logged in, on Plants page	1. Check page header/footer	No "Add Plant" button anywhere on page
TC_UI_USER_003	User can search plants	Verify search works for users	User logged in, plant "Orchid" exists	1. Type "Orchid" in search 2. Press Enter	Matching plants displayed Search results clear
TC_UI_USER_004	User can filter by category	Verify category filter for users	User logged in, multiple categories	1. Select "Succulents" from filter	Filtered plants displayed Only view access
TC_UI_USER_005	User can sort plants	Verify sorting for users	User logged in, multiple plants	1. Click "Name" column header	Plants sorted alphabetically Sort indicators visible
TC_UI_USER_006	User cannot edit/delete plants	Verify read-only access	User logged in, on plant list	1. Hover over plant rows 2. Right-click on plants 3. Check context menu	No edit/delete icons No edit options in menu
TC_UI_USER_007	User can view plant details	Verify detail view access	User logged in, plant "Cactus" exists	1. Click on Cactus plant name 2. View details page	Plant details page opens All info visible No edit form
TC_UI_USER_008	User UI is responsive	Verify mobile responsiveness	User logged in, on mobile device	1. View plants page 2. Rotate device 3. Test filters/search	Layout adapts to screen size All functions work on mobile No horizontal scrolling
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UI TEST CASES – ADMIN ROLE					

TC_UI_ADMIN_001	Admin sees Plant List with pagination	Verify plant list displays correctly	Admin logged in	1. Navigate to /admin/plants 2. Scroll down Plant table shows 10 plants/page	Pagination controls visible
TC_UI_ADMIN_002	Admin sees "Add Plant" button	Verify add button is visible	Admin logged in, on Plants page	1. Navigate to Plants page	"Add New Plant" button visible top-right
TC_UI_ADMIN_003	Admin can add new plant with valid data	Verify plant creation form works	Admin logged in, sub-categories exist	1. Click "Add Plant" 2. Fill valid data 3. Click Save	Success toast appears New plant visible in list
TC_UI_ADMIN_004	Admin can edit existing plant	Verify edit functionality	Admin logged in, plant "Rose" exists	1. Click edit icon on Rose 2. Update price 3. Click Save	Success message Updated price shows in list
TC_UI_ADMIN_005	Admin can delete plant with confirmation	Verify delete flow with confirmation	Admin logged in, plant "Tulip" exists	1. Click delete icon on Tulip 2. Confirm in dialog 3. Check list	Confirmation dialog appears Tulip removed from list
TC_UI_ADMIN_006	Admin sees validation errors for invalid inputs	Verify form validation	Admin logged in, on Add Plant form	1. Enter name "AB" 2. Enter price "0" 3. Try to save	Red error messages under each field Save button disabled
TC_UI_ADMIN_007	Admin can search plants by name	Verify search functionality	Admin logged in, plant "Lily" exists	1. Type "Lily" in search box 2. Press Enter	2. Only plants with "Lily" in name shown Clear results
TC_UI_ADMIN_008	Admin can filter plants by category	Verify category filter	Admin logged in, multiple categories exist	1. Select "Indoor Plants" from filter dropdown	Only plants from selected category shown
TC_UI_ADMIN_009	Admin can sort plants by name/price/quantity	Verify sorting functionality	Admin logged in, multiple plants exist	1. Click "Price" column header 2. Click againSorts ascending on first click	Sorts descending on second click

TC_UI_ADMIN_010	Admin sees "No plants found" message	Verify empty state	Admin logged in, no plants match search	1. Search for "XYZ123" 2. Apply impossible filter	"No plants found" message shown Suggestions to clear filters
TC_UI_ADMIN_011	Admin plant form cancel button works	Verify cancel functionality	Admin logged in, on Add Plant form	1. Enter some data 2. Click Cancel 3. Confirm	Returns to plant list Unsaved data discarded
TC_UI_ADMIN_012	Admin plant list updates after actions	Verify real-time updates	Admin logged in, plant list open	1. Add new plant in another tab 2. Return to list tab	Plant list auto-refreshes New plant appears without reload
TC_UI_ADMIN_013	Admin can only select sub-categories in dropdown	Verify dropdown restrictions	Admin logged in, on Add Plant form	1. Click category dropdown 2. Try to select main category	Only sub-categories shown in dropdown Main categories disabled/grayed
TC_UI_ADMIN_014	Admin sees real-time stock updates	Verify stock changes reflect immediately	Admin logged in, two tabs open	1. Tab1: View plant with qty=10 2. Tab2: Update qty to 5 3. Return to Tab1	Tab1 automatically updates to show qty=5
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API TEST CASES – USER ROLE					
TC_API_USER_001	User fetches all plants → 200	Verify users can view all plants	1. Valid user token 2. Plants exist in DB	2. 1. Send GET /api/plants 2. Add user auth headerStatus 200 OK	Returns plant list (read-only fields)
TC_API_USER_002	User fetches plant by ID → 200	Verify users can view specific plant	1. User token 2. Plant ID 5 exists	1. Send GET /api/plants/5 2. Add user header	Status 200 OK Returns plant details
TC_API_USER_003	User fetches plants by category → 200	Verify category filter for users	1. User token 2. Category 2 has plants	1. Send GET /api/plants?categoryId=2	Status 200 Returns filtered plants
TC_API_USER_004	User searches plants by name → 200	Verify search for users	1. User token 2. Plant "Rose" exists	1. Send GET /api/plants?search=Rose	Status 200 Returns matching plants

TC_API_USER_005	User paginates plants → 200	Verify pagination for users	1. User token 2. >15 plants exist	1. Send GET /api/plants?page=1&limit=5	Status 200 Returns first 5 plants
TC_API_USER_006	User tries to create plant → 403	Verify users cannot create	1. User token 2. Valid plant data	1. Send POST /api/plants with data	Status 403 Forbidden Error: "Unauthorized"
TC_API_USER_007	User tries to update plant → 403	Verify users cannot update	1. User token 2. Plant ID 8 exists	1. Send PUT /api/plants/8 with updates	Status 403 Forbidden Error: "Unauthorized"
TC_API_USER_008	User tries to delete plant → 403	Verify users cannot delete	1. User token 2. Plant ID 12 exists	1. Send DELETE /api/plants/12Status	403 Forbidden Error: "Unauthorized"
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API TEST CASES – ADMIN ROLE					
TC_API_ADMIN_001	Admin creates plant under valid sub-category → 201	Verify admin can successfully create a new plant	1. Valid admin token 2. Sub-category ID exists in DB	1. Send POST /api/plants 2. Include valid plant data 3. Add admin auth header	Status 201 Created Response contains plant ID & details
TC_API_ADMIN_002	Admin updates plant details → 200	Verify admin can update existing plant	1. Admin token 2. Plant ID 10 exists	1. Send PUT /api/plants/10 2. Include updated data 3. Add admin header	Status 200 OK Response shows updated plant data
TC_API_ADMIN_003	Admin deletes plant → 204	Verify admin can delete plant	1. Admin token 2. Plant ID 15 exists	1. Send DELETE /api/plants/15 2. Add admin header	Status 204 No Content GET /plants/15 returns 404
TC_API_ADMIN_004	Admin fetches plant by ID → 200	Verify admin can retrieve specific plant	1. Admin token 2. Plant ID 20 exists	1. Send GET /api/plants/20 2. Add admin header	Status 200 OK Response contains plant details
TC_API_ADMIN_005	Admin creates plant with negative quantity → 400	Verify quantity validation works	1. Admin token 2. Valid category exists	1. Send POST /api/plants 2. Set quantity to -5	Status 400 Error: "Quantity cannot be negative"

TC_API_ADMIN_006	Admin creates plant with name < 3 chars → 400	Verify name length validation	1. Admin token 2. Valid category exists	1. Send POST /api/plants 2. Set name to "AB	Status 400 Error: "Name must be 3-25 characters"
TC_API_ADMIN_007	Admin creates plant with price ≤ 0 → 400	Verify price validation works	1. Admin token 2. Valid category exists	1. Send POST /api/plants 2. Set price to 0 or negative	Status 400 Error: "Price must be > 0"
TC_API_ADMIN_008	Admin updates plant with invalid categoryId → 400	Verify category validation on update	1. Admin token 2. Plant ID 25 exists 3. Category 999 doesn't exist	1. Send PUT /api/plants/25 2. Set categoryId to 999	Status 400 Error: "Invalid category ID"
TC_API_ADMIN_009	Admin fetches non-existent plant → 404	Verify error for missing plant	1. Admin token 2. Plant ID 999 doesn't exist	1. Send GET /api/plants/999 2. Add admin header	Status 404 Not Found Error: "Plant not found"
TC_API_ADMIN_010	Admin fetches plants by category → 200	Verify category filtering works	1. Admin token 2. Category 3 has plants	1. Send GET /api/plants?categoryId=3	Status 200 Response contains only plants from category 3
TC_API_ADMIN_011	Admin searches plants by name → 200	Verify search functionality	1. Admin token 2. Plant "Tulip" exists	1. Send GET /api/plants?search=Tulip	Status 200 Response contains plants with "Tulip" in name
TC_API_ADMIN_012	Admin paginates plants with filters → 200	Verify pagination with filters	1. Admin token 2. >20 plants exist	1. Send GET /api/plants?page=2&limit=10&categoryId=3	Status 200 Response has page 2 data, 10 items max
TC_API_ADMIN_013	Admin fetches plant summary → 200	Verify summary/stats endpoint	1. Admin token 2. Plants exist in DB	1. Send GET /api/plants/summary	Status 200 Response contains counts, low stock alerts